Complex Bilinear Interaction Fields in a Physical Optical Scattering Engine

Qiushi Engine

Abstract

Bilinear interactions between paired inputs are central to modern multimodal learning, yet their physical realization remains largely unexplored. Here we demonstrate that a self-referenced optical scattering platform can natively measure a *complex bilinear interaction field* $\boldsymbol{B}(x,y) = \left(\overline{(TE_x)_m}\,(TE_y)_m\right)_{m\in\mathcal{S}}$, where T is the fixed transmission operator of a complex scattering medium and E_x, E_y are optical fields encoding two input tokens. Through four-phase interferometric demodulation and blank-subtraction calibration, we isolate this complex cross-term from raw intensity measurements on a real physical platform.

We report two physical experiments totalling 1,944 calibrated measurements. In a four-token XOR showcase, the measured complex interaction field simultaneously solves both pair identification (accuracy 1.000) and a parity relation that is inaccessible to separate-token linear readouts under the tested evaluation protocol (accuracy 1.000), whereas concatenated single-token features solve only identity and a matched-budget digital bilinear baseline solves only the relation. In a broader eight-token semantic benchmark spanning four natural-language categories (animals, tools, fruits, vehicles) and all 64 ordered pairs, the same structural separation persists: Complex-$\boldsymbol{B}$ is the only tested representation that jointly encodes pair identity, semantic category relation, and category-pair structure, achieving near-perfect accuracy on all three tasks under matched feature budgets as low as $k = 50$. We also map an honest failure boundary: for category-level generalization to unseen token pairs, the digital bilinear baseline outperforms Complex-$\boldsymbol{B}$ (1.000 versus 0.625), revealing that the physical field's strength is joint encoding of concrete pair structure rather than abstract category transfer.

These results separate two distinct questions: Complex-$\boldsymbol{B}$ provides a richer measured pair representation that exposes interaction structure unavailable to separate-token linear features, but this expressive advantage does not imply superior category-level extrapolation to unseen token pairs. Together, our experiments establish measurable complex interaction fields as a physically grounded form of bilinear computation and delineate the regime where such physical representations offer structural advantages over their digital counterparts.

1 Introduction

Bilinear interactions—multiplicative couplings between two input representations—are a cornerstone of modern machine learning. From visual question answering [1] to multimodal fake-news detection [2], methods based on bilinear pooling, factorized bilinear models, and Hadamard-product fusion [3] have demonstrated that explicitly modelling pairwise interactions between heterogeneous inputs yields richer representations than linear concatenation alone. The central computational challenge is dimensionality: a full bilinear map between two d-dimensional vectors produces a d^2-dimensional output, motivating low-rank and factorized approximations that trade expressiveness for tractability.

In parallel, the optical-computing community has explored physical substrates that perform high-dimensional transformations at the speed of light. Complex scattering media—multimode fibres, diffusing layers, and random photonic structures—act as fixed, high-dimensional linear operators whose transmission matrices T map input optical fields to output speckle patterns [4]. Recent work has demonstrated that such media can serve as random-feature engines for classification [5], reservoir computers [6], and even programmable linear processors when combined with wavefront shaping. However, the vast majority of these approaches treat the scattering medium as a *single-input* device: one optical field enters, one intensity pattern exits.

A largely unexplored opportunity lies at the intersection of these two threads. When two optical fields E_x and E_y are coherently superposed and jointly propagated through a scattering medium, the detected intensity contains not only single-token contributions but also a *cross-term* that is physically bilinear in the two input fields. This cross-term is not an artefact to be suppressed; it is a measurable complex interaction field

$$B_m(x, y) = \overline{(TE_x)_m}\,(TE_y)_m, \tag{1}$$

accessible through phase-stepped interferometric demodulation. The key insight is that the scattering medium, combined with coherent detection, *natively computes* a structured bilinear interaction between two inputs—without any digital multiplication, learned projection, or explicit outer-product construction.

Contribution. This paper establishes the measurable complex interaction field $\boldsymbol{B}(x, y)$ as the primary physical object of a bilinear optical interaction engine and provides rigorous experimental evidence for its computational role. Our contributions are:

1. **Theory and object discipline.** We derive $\boldsymbol{B}(x, y)$ from the coherent superposition and square-law detection model, isolate it via four-phase demodulation and blank subtraction, and enforce a strict discipline: the complex field itself—not its magnitude, real part, or imaginary part—is the sole research object (section 2).

2. **XOR showcase experiment.** On a four-token library, we demonstrate that $\boldsymbol{B}$ simultaneously solves pair identification and a parity/XOR relation that is inaccessible to separate-token linear readouts under the tested linear evaluation protocol, while matched baselines solve at most one task each (section 3.1).

3. **Semantic benchmark experiment.** On an eight-token library with four natural-language semantic categories, we show that the same structural separation scales: $\boldsymbol{B}$ is the only tested representation jointly encoding pair identity, category relation, and category-pair structure under matched feature budgets (section 3.2).

4. **Failure-boundary mapping.** We identify and report an honest limitation: for category-level generalization to unseen token pairs, a simple digital bilinear baseline outperforms the physical field, delineating the regime where physical bilinear computation offers structural advantages (section 3.4).

Importantly, our results separate two questions that are often conflated: the *expressive advantage* of a representation (what pair structure it makes linearly accessible) and its *generalization advantage* (whether that structure transfers to unseen inputs). Complex-$\boldsymbol{B}$ excels at the former but not unconditionally at the latter, and we report both findings with equal emphasis. Together, these results position complex scattering media not merely as random-feature generators or single-input embedding engines, but as *physical bilinear interaction engines* whose native observables carry structured pairwise information.

2 Theory: Complex Interaction Field as a Physical Bilinear Observable

2.1 Coherent superposition and square-law detection

Consider a fixed complex scattering medium with transmission operator T mapping the spatial-light-modulator (SLM) plane to the detector plane. For a single input token x encoded as optical field E_x, the output field at detector channel m is $(TE_x)_m$. The platform employs a static self-reference field s_m that co-propagates through the medium.

When two tokens x and y are encoded simultaneously with a controlled phase offset ϕ on the second input, the total field at channel m is

$$F_m^{(\phi)}(x, y) = s_m + (TE_x)_m + e^{i\phi}(TE_y)_m. \tag{2}$$

The camera records intensity $I_m^{(\phi)}(x, y) = |F_m^{(\phi)}(x, y)|^2$.

2.2 Four-phase demodulation

Using the four phase settings $\phi \in \{0,\, \pi/2,\, \pi,\, 3\pi/2\}$, we define the demodulated complex quantity

$$Q_m(x,y) = \frac{I_m^{(0)} - I_m^{(\pi)}}{4} + i\,\frac{I_m^{(3\pi/2)} - I_m^{(\pi/2)}}{4}. \tag{3}$$

Direct algebraic expansion of (2) yields

$$Q_m(x,y) = \overline{s_m}\,(TE_y)_m + \overline{(TE_x)_m}\,(TE_y)_m. \tag{4}$$

The first term depends only on the reference and on y; the second is the bilinear cross-term of interest.

2.3 Blank subtraction: isolating the interaction field

To remove the reference-dependent contribution, we measure a *blank control* with $x = 0$ (zero field on the x-input channel):

$$Q_m(0,y) = \overline{s_m}\,(TE_y)_m. \tag{5}$$

Subtracting yields the *complex interaction field*:

$$\boxed{B_m(x,y) = Q_m(x,y) - Q_m(0,y) = \overline{(TE_x)_m}\,(TE_y)_m.} \tag{6}$$

Collecting over the selected detector channels $\mathcal{S}$ gives the complex interaction vector

$$\boldsymbol{B}(x,y) = \big(B_m(x,y)\big)_{m\in\mathcal{S}} \;\in\; \mathbb{C}^{|\mathcal{S}|}. \tag{7}$$

Definition 2.1 (Object discipline)**.** The primary physical object of this work is the complex vector $\boldsymbol{B}(x,y)$. For numerical computation, we use the real-coordinate representation $\rho(\boldsymbol{B}) = [\mathrm{Re}\,\boldsymbol{B};\ \mathrm{Im}\,\boldsymbol{B}] \in \mathbb{R}^{2|\mathcal{S}|}$, which is a lossless bijection preserving all information. The derived quantities $|\boldsymbol{B}|$, $\mathrm{Re}\,\boldsymbol{B}$, and $\mathrm{Im}\,\boldsymbol{B}$ taken individually are *not* the research object: they discard phase information or project the complex field onto a reduced observable.

2.4 Physical bilinearity of the interaction field

Equation (6) shows that $B_m(x,y)$ is the product of two linearly transformed inputs:

$$B_m(x,y) = \sum_j \overline{T_{mj}}\,\overline{(E_x)_j}\;\cdot\;\sum_k T_{mk}\,(E_y)_k. \tag{8}$$

This is a *bilinear* function of (E_x, E_y): linear in each argument when the other is held fixed. The scattering medium acts as a high-dimensional, fixed, physically determined

bilinear kernel—analogous to the weight matrices in factorized bilinear pooling [3], but implemented by wave propagation through a disordered photonic structure rather than by learned digital parameters.

2.5 Comparison with digital bilinear baselines

Given single-token embeddings $z(x) = (|(TE_x)_m|^2)_{m\in\mathcal{S}}$ obtained from intensity-only measurements, one can construct a *digital elementwise bilinear* feature:

$$d_m(x, y) = z_m(x) \cdot z_m(y) = |(TE_x)_m|^2 \, |(TE_y)_m|^2. \tag{9}$$

This is a product of *intensities* (real, non-negative), whereas the physical $B_m(x, y)$ is a product of *complex amplitudes* (carrying phase). The digital bilinear baseline therefore discards all relative-phase information between the two tokens' scattered fields. This distinction is central to the experimental results that follow.

2.6 Task definitions

We preregister two classes of tasks that probe different aspects of the interaction field:

Identity tasks. Given $\boldsymbol{B}(x, y)$ or a baseline feature, identify the ordered pair (x, y). This tests whether the representation preserves token-specific information within the interaction.

Relation tasks. Given $\boldsymbol{B}(x, y)$ or a baseline feature, classify a *relation* between x and y that depends on their joint properties (e.g., parity, semantic category membership) rather than on individual token identity. By the XOR/parity theorem, certain relation labels—specifically, those forming a checkerboard pattern over the token index space—are linearly inseparable from concatenated single-token features $[z(x), z(y)]$, regardless of the dimensionality of z.

3 Results

3.1 Experiment 1: XOR showcase (4 tokens, 432 measurements)

Setup. Four phase-only tokens are encoded on the SLM and measured through the scattering platform using the four-phase demodulation protocol described in section 2. All 16 ordered pairs are measured with 5 repeats per phase setting, plus blank controls and drift checks, yielding 432 calibrated measurements in 30.7 minutes.

Metrology validation. The extracted complex interaction field $\boldsymbol{B}(x, y)$ shows excellent repeat-to-repeat stability: the mean Pearson correlation of the real-coordinate representation $\rho(\boldsymbol{B})$ across repeat pairs is 0.9875, with a minimum of 0.9589 (fig. 4). Representative four-phase intensity traces (fig. 1) show modulation at selected detector channels consistent with the expected sinusoidal dependence on ϕ. Figure 2 shows the extracted complex field $\boldsymbol{B}(x, y)$ for four representative pairs plotted in the complex plane, revealing structured, pair-specific channel distributions.

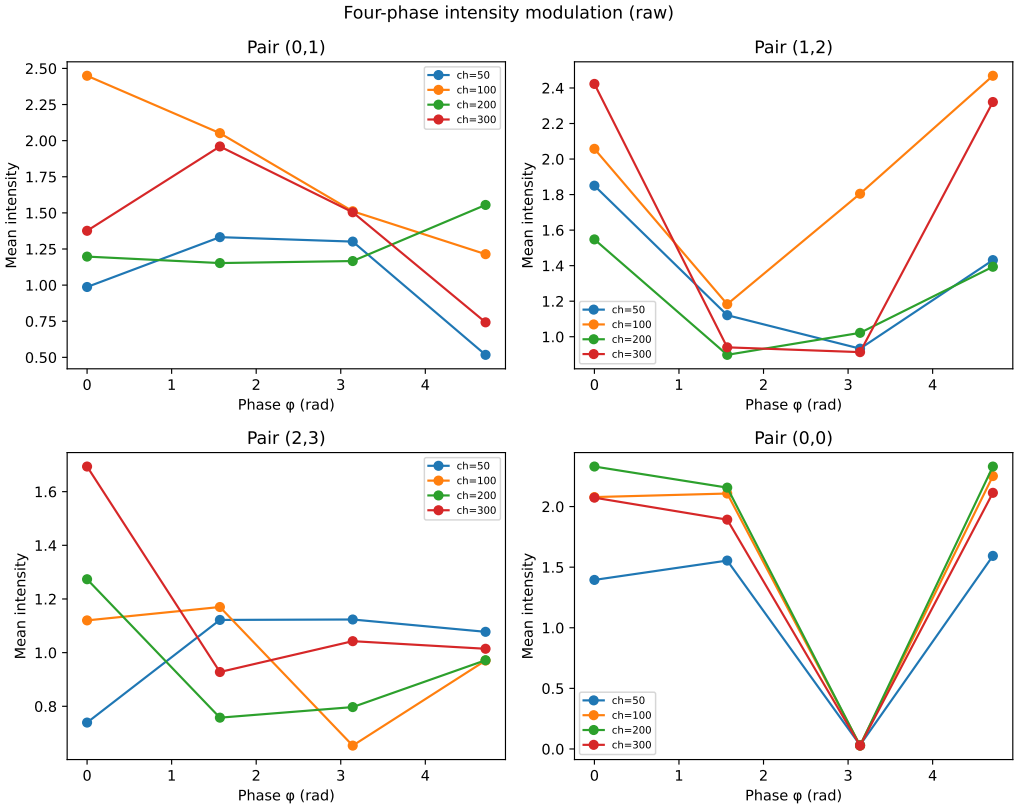


Figure 1: Four-phase intensity modulation at selected detector channels for four representative token pairs. The measured intensities are consistent with the expected sinusoidal dependence on ϕ assumed in the demodulation (eq. (3)). Note that four phase samples cannot independently validate a sinusoidal model; these traces illustrate consistency rather than provide a stringent test.

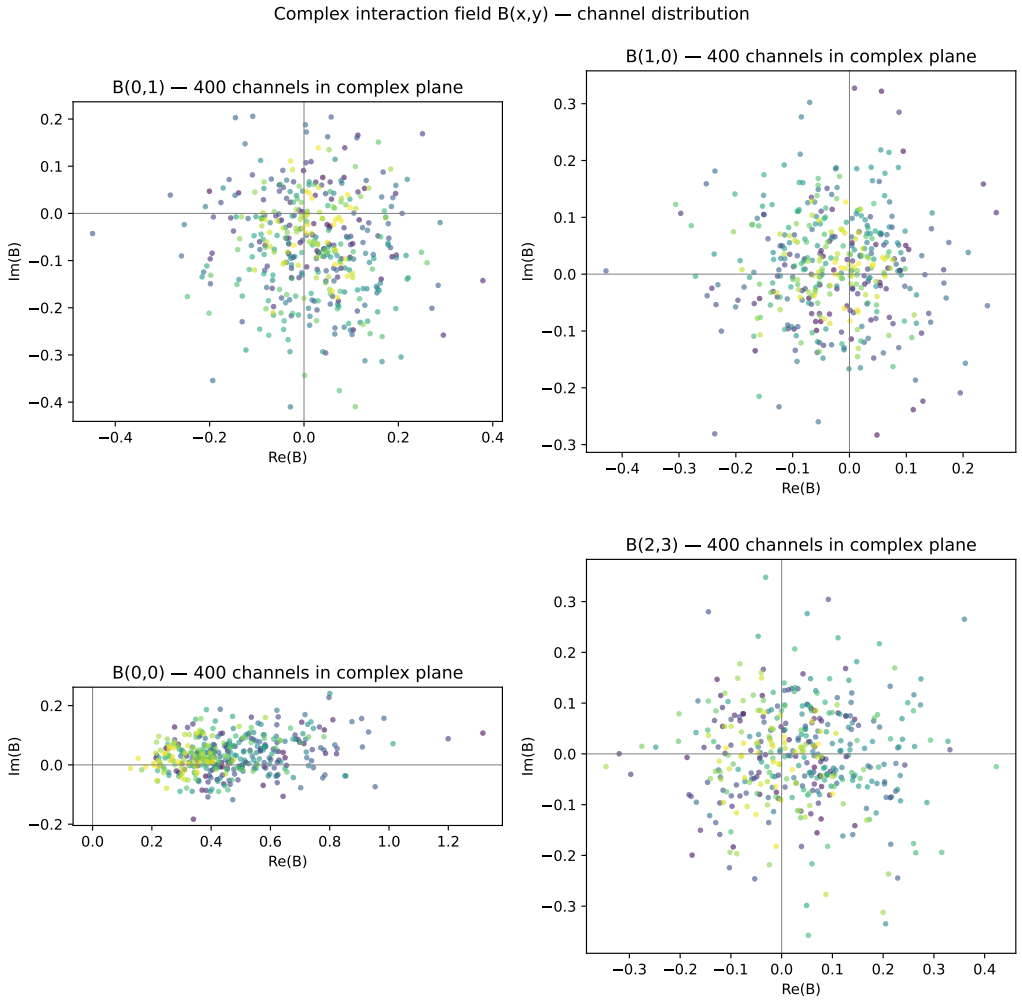


Figure 2: Complex interaction field $\boldsymbol{B}(x, y)$ for four token pairs, with each detector channel plotted as a point in the complex plane. Different pairs produce distinct, structured distributions, demonstrating that the interaction field carries pair-specific information.

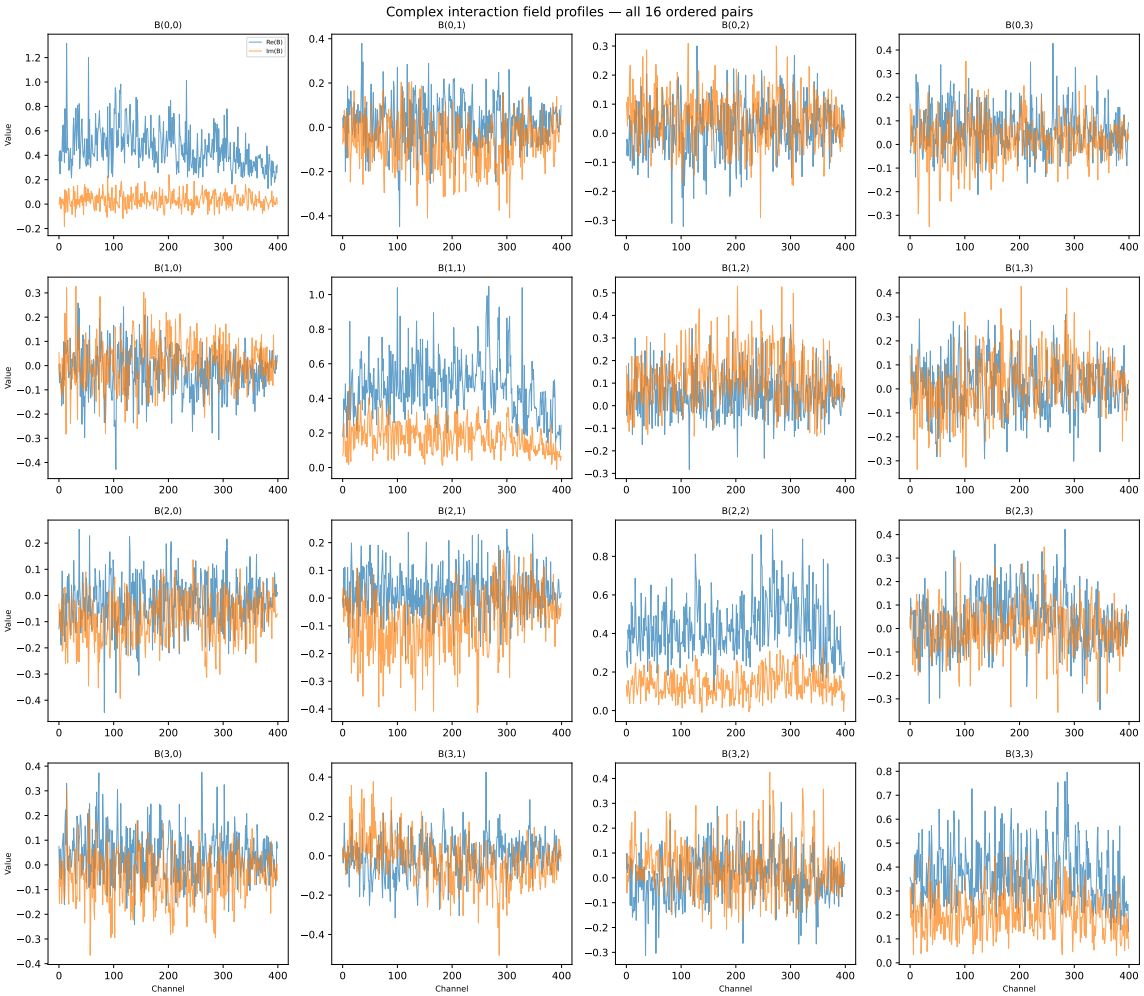


Figure 3: Real and imaginary parts of $\boldsymbol{B}(x, y)$ across all 400 detector channels for all 16 ordered pairs. Each subplot corresponds to one pair (x, y). The diversity of channel profiles across pairs underlies the high pair-identification accuracy.

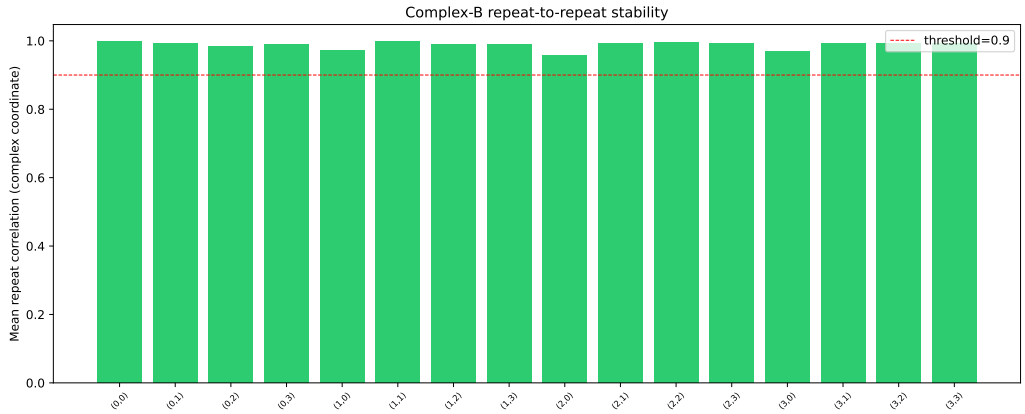


Figure 4: Repeat-to-repeat stability of the complex interaction field. Each bar shows the mean Pearson correlation of $\rho(\boldsymbol{B})$ across all repeat pairs for one ordered token pair. Mean correlation: 0.9875; minimum: 0.9589.

XOR task definition. We define a binary parity label: $\ell(x, y) = (x + y) \bmod 2$. Same-parity pairs (e.g., $(0, 0)$, $(1, 3)$) receive label 0; cross-parity pairs (e.g., $(0, 1)$, $(2, 3)$) receive label 1. This label forms a checkerboard pattern in the token-index space and is linearly inseparable from any concatenation $[z(x), z(y)]$ of single-token features, regardless of the dimensionality of z.

Classification results. Table 1 reports leave-one-out cross-validated balanced accuracy for eight feature families on both pair identification (Task A, 16 classes) and XOR parity (Task B, binary).

Table 1: XOR showcase results (LOO-CV balanced accuracy). Complex-$\boldsymbol{B}$ is the only feature family solving both tasks simultaneously.

Feature family	Task A (pair ID)	Task B (XOR)
$z(x)$ only	0.000	0.000
$z(y)$ only	0.000	0.000
$[z(x), z(y)]$ concat	**1.000**	0.000
$z(x) + z(y)$	0.250	0.000
$z(x) - z(y)$	0.750	0.000
Digital bilinear $z(x) \odot z(y)$	0.250	**1.000**
Complex-$\boldsymbol{B}$ $[\mathrm{Re}\,\boldsymbol{B};\ \mathrm{Im}\,\boldsymbol{B}]$	**1.000**	**1.000**
Hybrid $[z(x), z(y), \mathrm{Re}\,\boldsymbol{B}, \mathrm{Im}\,\boldsymbol{B}]$	1.000	1.000

The result exhibits a clean three-way structural separation:

- **Concatenation** solves identity but completely fails the XOR relation (0.000).
- **Digital bilinear** solves the XOR relation but fails identity (0.250).
- **Complex-$\boldsymbol{B}$** solves both simultaneously (1.000 / 1.000).

Figure 5 visualizes this separation, and fig. 6 shows the XOR class structure in a two-dimensional PCA projection of the Complex-$\boldsymbol{B}$ space.

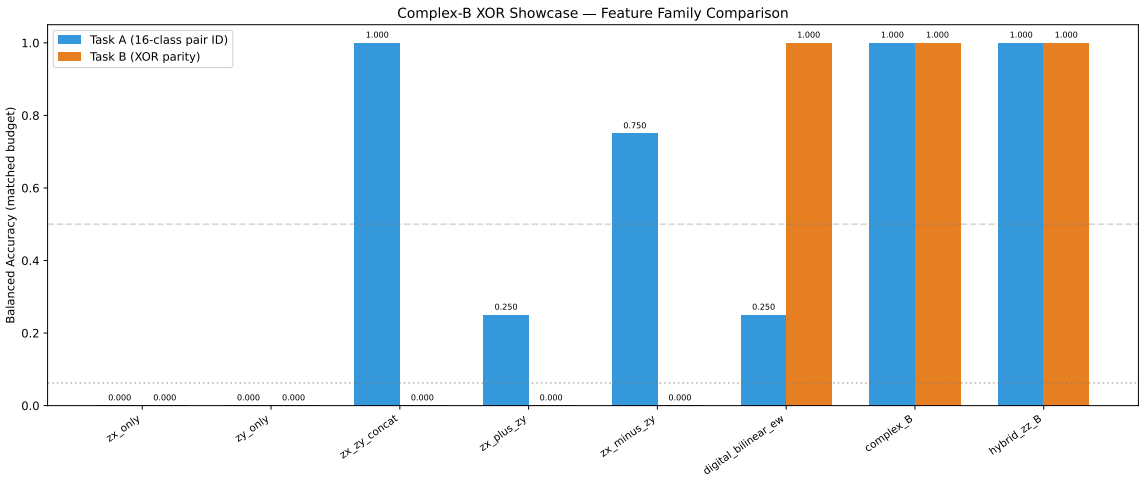


Figure 5: Feature-family comparison on the XOR showcase. Complex-$\boldsymbol{B}$ (red star) is the only family achieving high accuracy on both tasks. Concatenation fails XOR; digital bilinear fails pair identification.

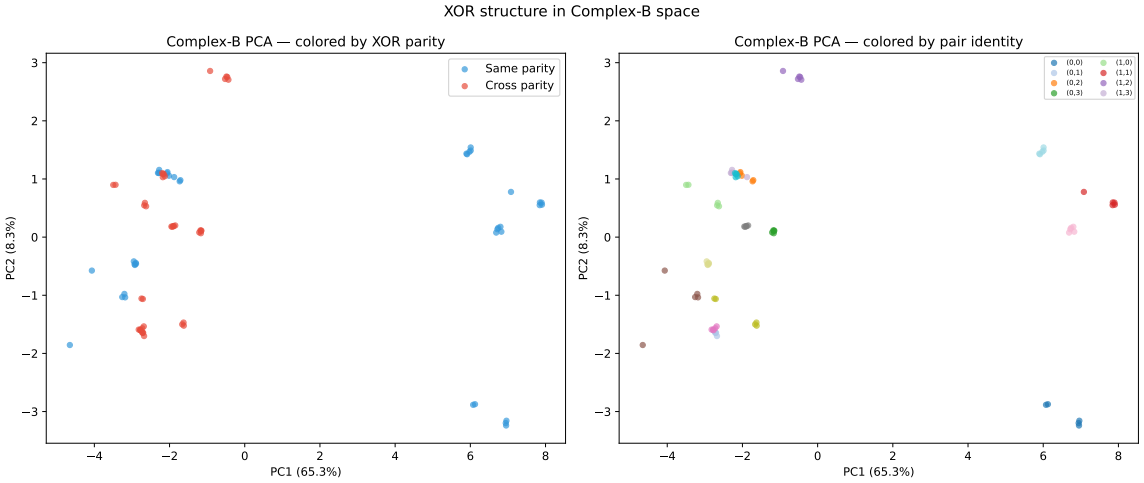


Figure 6: PCA projection of Complex-$\boldsymbol{B}$ samples. **Left:** colored by XOR parity label, showing clear two-class separation. **Right:** colored by pair identity, showing 16 distinct clusters. The interaction field simultaneously encodes both structures.

3.2 Experiment 2: Semantic benchmark (8 tokens, 1,512 measurements)

Setup. Eight semantic tokens from four natural-language categories are encoded as deterministic phase-only SLM fields: *animals* (cat, dog), *tools* (hammer, saw), *fruits* (apple, grape), *vehicles* (car, train). Each token maps directly to a unique optical field via a documented, deterministic superpixel codebook (seed $= 10{,}000 + \text{id} \times 7919$), preserving a true dual-input structure with no intermediate random projection. All 64 ordered pairs are measured with the full four-phase protocol, yielding 1,512 calibrated measurements in 93.3 minutes. CMOS1 documentation shots record the generated token fields (fig. 7).

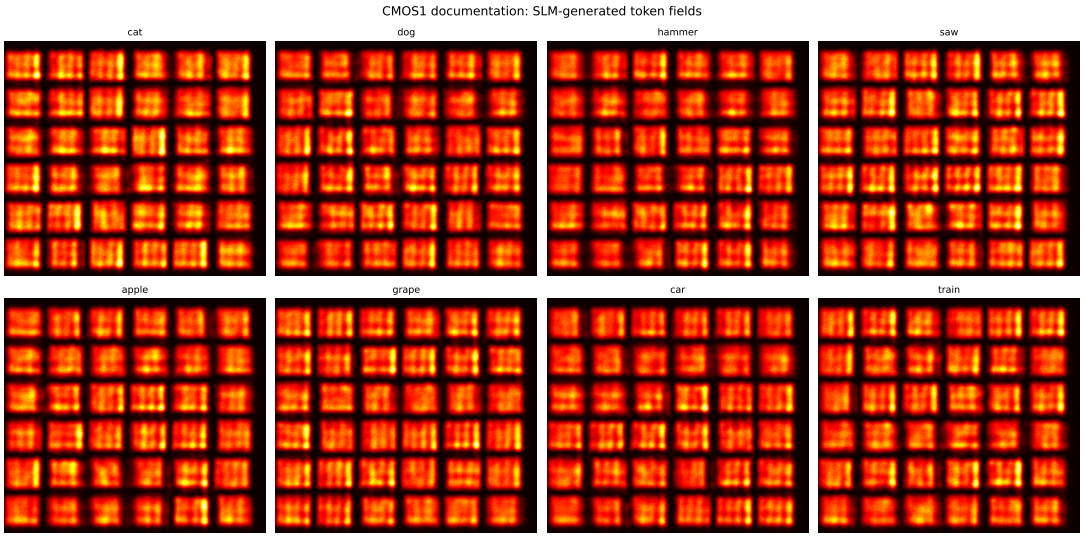


Figure 7: CMOS1 documentation images of the eight semantic token fields generated on the SLM. Each token produces a distinct speckle-like intensity pattern after propagation, confirming that the deterministic phase-only codebook generates physically distinguishable fields.

Phase-cycle validation. Representative four-phase intensity traces for semantic pairs show modulation consistent with the expected sinusoidal model (fig. 8), extending the pattern observed in Experiment 1 to the eight-token scale.

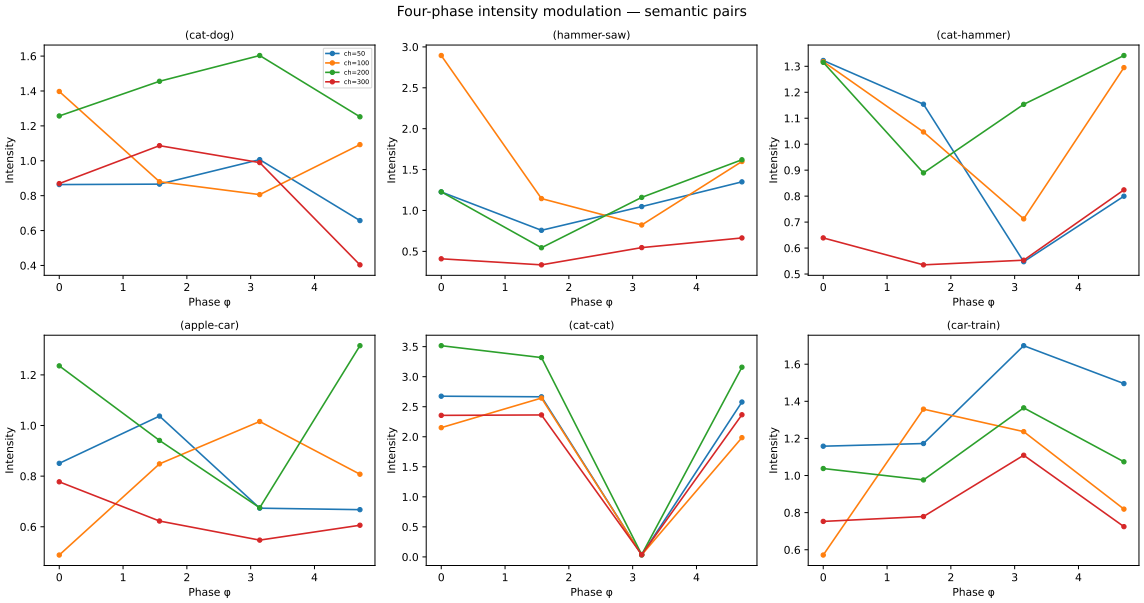


Figure 8: Four-phase intensity modulation for representative semantic token pairs, including same-category pairs (cat–dog, hammer–saw) and cross-category pairs (cat–hammer, apple–car). The measured intensities are consistent with the sinusoidal dependence on ϕ assumed by the demodulation protocol. As in fig. 1, four phase samples illustrate consistency rather than provide an independent model test.

Stability. Complex-$\boldsymbol{B}$ repeat stability across all 64 pairs: mean correlation 0.9828, minimum 0.8567 (fig. 9). Within-group repeat correlations for singles and pairs exceed 0.995, confirming platform stability throughout the 93-minute acquisition.

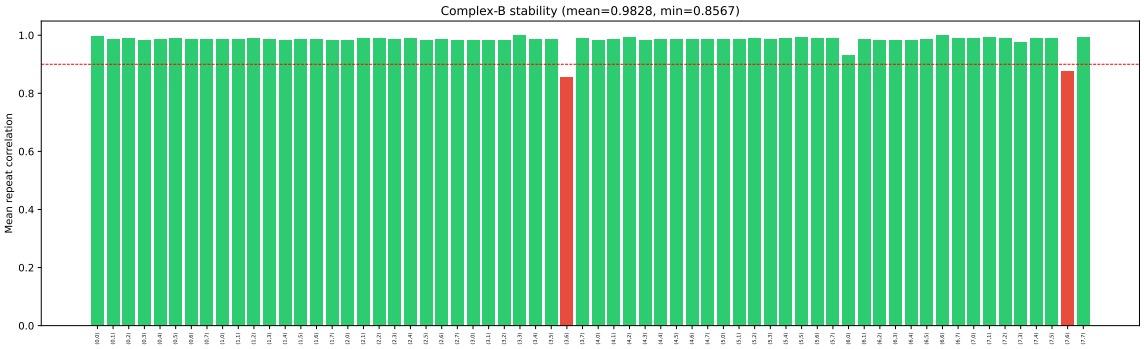


Figure 9: Complex-$\boldsymbol{B}$ repeat stability for all 64 ordered semantic pairs. Mean: 0.9828; minimum: 0.8567. The platform maintains adequate stability for reliable interaction-field extraction.

Semantic tasks. Three preregistered tasks probe different levels of pair structure:

- **Task A**: 64-class ordered pair identification.
- **Task B**: Binary same-category vs. different-category classification. The classifier receives only the pair representation, not the category identities themselves; the task tests whether category agreement is linearly recoverable from the representation alone.
- **Task C**: 16-class category-pair identification (which category does x belong to, and which does y belong to?).

Classification results. Table 2 reports LOO-CV balanced accuracy.

Table 2: Semantic benchmark results (LOO-CV balanced accuracy). Complex-$\boldsymbol{B}$ is the only feature family solving all three tasks simultaneously.

Feature family	Task A (pair ID)	Task B (same-category)	Task C (category-pair)
$z(x)$ only	0.000	0.500	0.000
$z(y)$ only	0.000	0.500	0.000
$[z(x),\ z(y)]$ concat	**1.000**	0.500	**1.000**
$z(x) + z(y)$	0.125	0.500	0.250
$z(x) - z(y)$	0.875	0.500	0.750
Digital bilinear $z(x) \odot z(y)$	0.125	**1.000**	0.250
Complex-$\boldsymbol{B}$	**1.000**	**1.000**	**1.000**
Hybrid	1.000	1.000	1.000

The three-way structural separation from the XOR showcase replicates at semantic scale: concatenation solves identity tasks (A, C) but fails the relational task B at chance level (0.500); digital bilinear solves the relational task but fails identity; Complex-$\boldsymbol{B}$ solves all three.

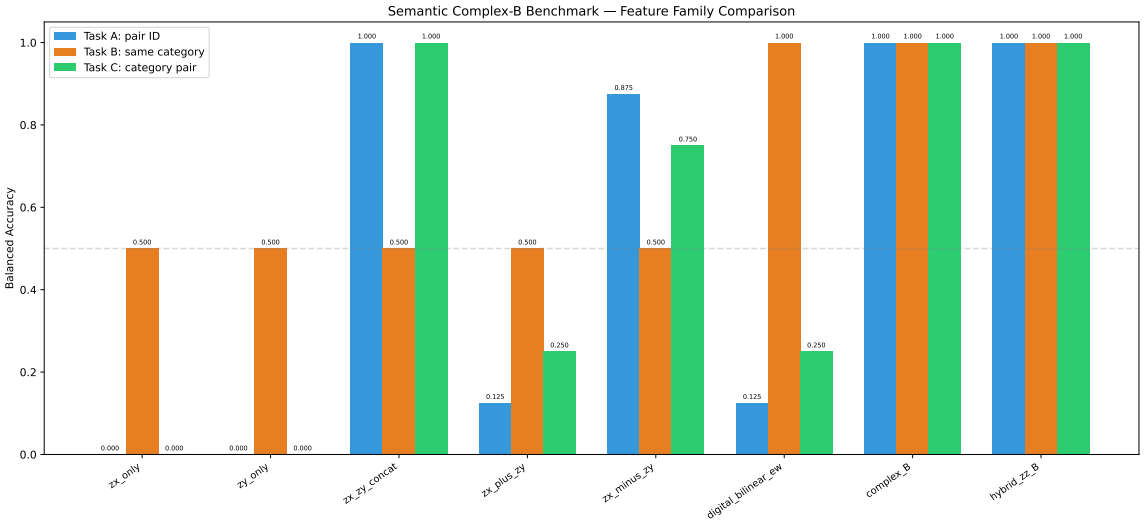


Figure 10: Semantic benchmark comparison across all feature families and three tasks. Complex-$\boldsymbol{B}$ achieves perfect or near-perfect accuracy on all tasks; no other single family matches this.

Figure 11 contrasts the complex-plane distributions of $\boldsymbol{B}$ for same-category and cross-category pairs, and fig. 12 shows PCA projections colored by category structure.

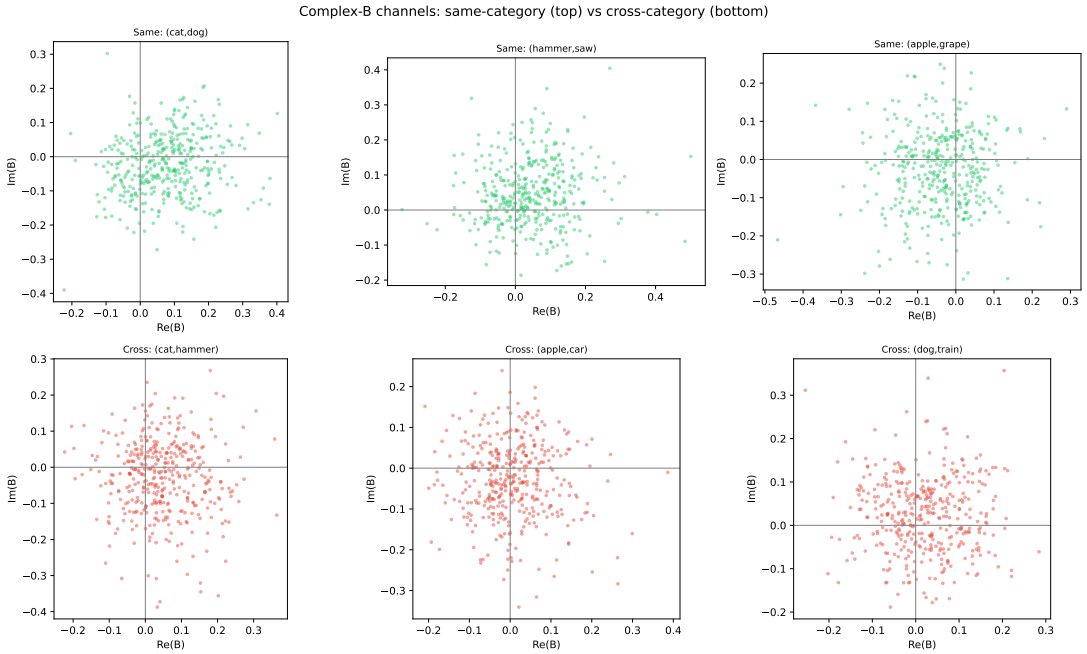


Figure 11: Complex-$\boldsymbol{B}$ channel distributions for same-category pairs (top, green) and cross-category pairs (bottom, red). The interaction fields show visually distinct structures, consistent with the perfect same-category classification.

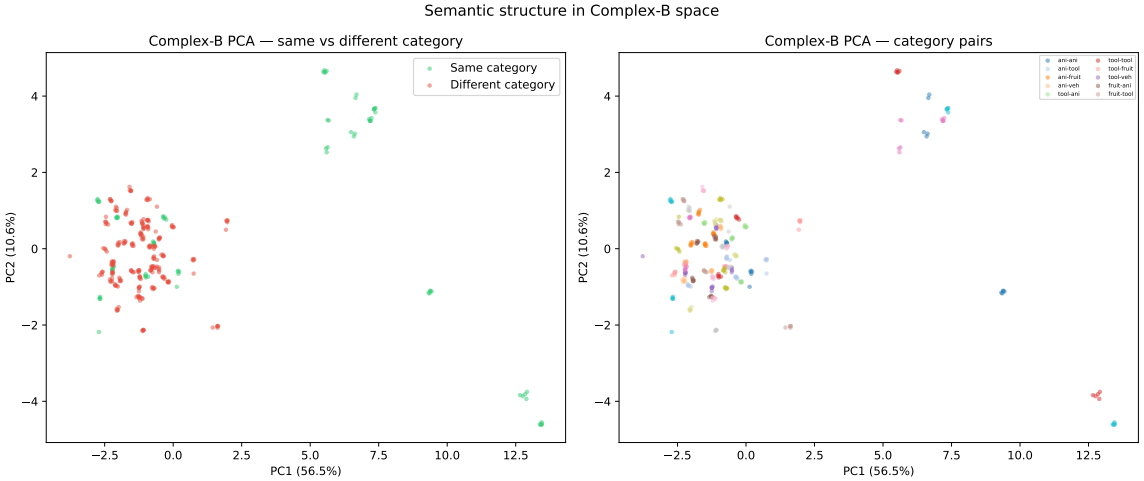


Figure 12: PCA projection of Complex-$\boldsymbol{B}$ samples. **Left:** colored by same/different category. **Right:** colored by category pair. Semantic structure is visible in the first two principal components.

3.3 Compressed-budget fairness analysis

Comparison principle. The three key comparison families—concatenation $[z(x), z(y)]$, digital bilinear $z(x) \odot z(y)$, and Complex-$\boldsymbol{B}$—are chosen to isolate the role of coherent phase information. Concatenation preserves both tokens' full intensity embeddings but forms no interaction; digital bilinear forms an elementwise interaction but uses only intensity (discarding phase); Complex-$\boldsymbol{B}$ forms a physical interaction that retains coherent phase. In table 2, feature dimensionalities differ across families (400–1,600). To control for this, we additionally evaluate all families at *matched* feature budgets $k \in \{10, 25, 50, 100, 200, 400\}$ using random-subspace selection averaged over 10 trials, ensuring that any observed separation reflects representation content rather than dimensionality.

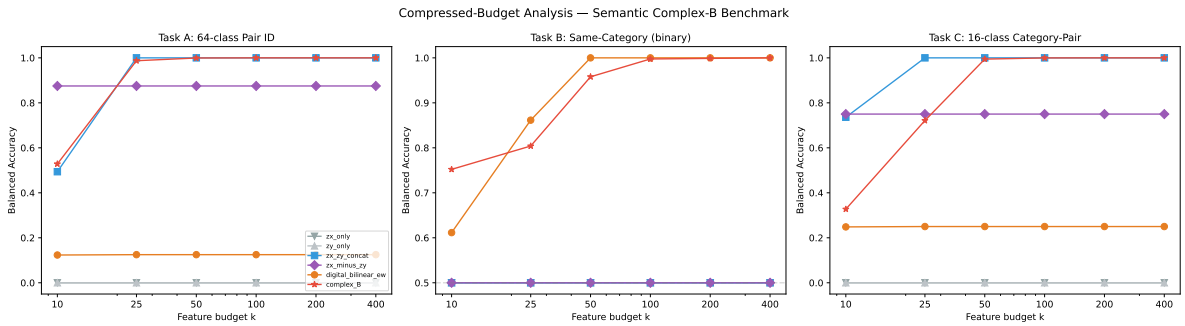


Figure 13: Compressed-budget analysis across three tasks. Complex-$\boldsymbol{B}$ (red stars) achieves joint encoding at every budget, while concatenation (blue squares) never solves the relational task and digital bilinear (orange circles) never solves identity, regardless of budget.

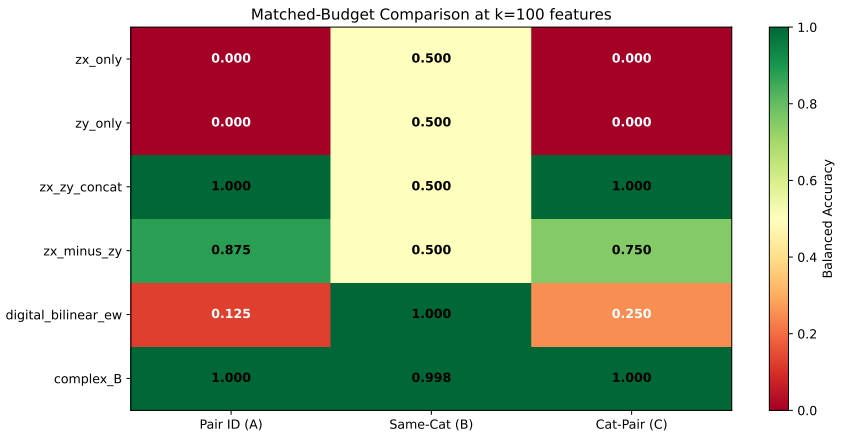


Figure 14: Matched-budget heatmap at $k = 100$ features. Complex-$\boldsymbol{B}$ is the only family achieving high accuracy on all three tasks simultaneously under equal feature budget.

At $k = 100$ (matched for all families), Complex-$\boldsymbol{B}$ achieves Task A = 1.000, Task B = 0.998, Task C = 1.000. Concatenation remains at Task B = 0.500 (chance) at *every* budget from 10 to 400. Digital bilinear remains at Task A = 0.125 at every budget. The structural separation is therefore *budget-independent*: it is a qualitative property of the representations, not a consequence of dimensionality mismatch.

3.4 Failure boundary: cross-split generalization

To test whether the category-relation signal in Complex-$\boldsymbol{B}$ generalizes to unseen token pairs, we hold out 16 ordered pairs (one per category-pair combination) and train on the remaining 48.

Table 3: Cross-split generalization on same-category classification (Task D). Digital bilinear generalizes perfectly; Complex-$\boldsymbol{B}$ does not.

Feature family	Cross-split accuracy	Train/Test split
$[z(x), z(y)]$ concat	0.500	240 / 80
Digital bilinear $z(x) \odot z(y)$	**1.000**	240 / 80
Complex-$\boldsymbol{B}$	0.625	240 / 80
Hybrid	0.458	240 / 80

The digital bilinear baseline achieves perfect generalization (1.000) because it operates on single-token intensity embeddings that are shared between training and test pairs: category structure learned from some pairs transfers directly. Complex-$\boldsymbol{B}$, by contrast, encodes pair-specific physical interactions that do not trivially transfer across unseen pair combinations, yielding only 0.625. Figure 15 visualizes the cross-split accuracy across all tested feature families.

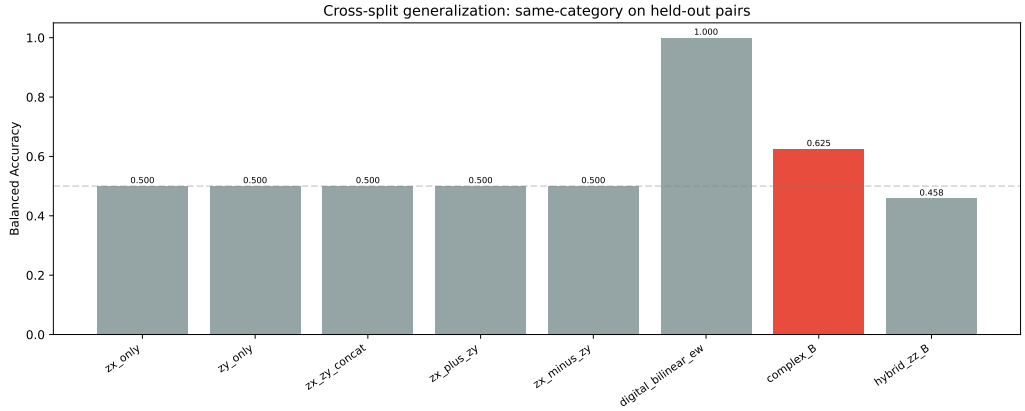


Figure 15: Cross-split generalization accuracy for same-category classification (Task D). The digital bilinear baseline generalizes perfectly to unseen token pairs, while Complex-$\boldsymbol{B}$ achieves only 0.625, illustrating the trade-off between pair-specific expressiveness and abstract category transfer.

This is not a failure of the measurement but a genuine structural property: the richness that makes Complex-$\boldsymbol{B}$ a powerful joint encoder for *known* pairs also makes it less suited for abstract category-level transfer from limited training data. We report this boundary honestly as a scientifically informative result.

Statistical note. All reported accuracies are leave-one-out cross-validated point estimates. We do not report bootstrap confidence intervals; however, the central structural claims—that concatenation *never* solves the relational task (0.500 at all budgets) and digital bilinear *never* solves identity (0.125 at all budgets)—are qualitative separations at floor or chance level, not marginal differences that depend on statistical power.

4 Discussion

4.1 Complex interaction fields as physical bilinear representations

The central finding of this work is that a self-referenced optical scattering platform natively produces a measurable complex bilinear interaction field $\boldsymbol{B}(x, y)$ that jointly encodes pair identity and pair relation—two information axes that are structurally separated across all tested digital baselines. This joint-encoding property is not a marginal quantitative improvement; it is a qualitative structural difference that persists across two experiments, two vocabulary scales, multiple task types, and all tested feature budgets.

The physical origin of this property is the coherent cross-term $\overline{(TE_x)_m}(TE_y)_m$, which preserves relative-phase information between the two tokens' scattered fields. The digital bilinear baseline $z(x) \odot z(y) = |(TE_x)_m|^2 |(TE_y)_m|^2$ discards this phase, retaining only

intensity-level correlations. Our results suggest that the phase-carrying cross-term is precisely what enables simultaneous identity and relation encoding: identity information is carried by the pair-specific amplitude and phase profile of $\boldsymbol{B}$, while relation information is carried by systematic structural similarities across pairs sharing the same relation label.

It is important to distinguish two properties that are often conflated. The *expressive advantage* of Complex-$\boldsymbol{B}$—its ability to make both identity and relation linearly accessible from a single representation—is strongly supported by both experiments. The *generalization advantage*—whether category-level structure transfers to unseen token pairs—is *not* established: the cross-split experiment (section 3.4) shows that digital bilinear features generalize better for abstract category transfer. These are complementary findings, not contradictory ones: a representation can be expressively rich for known inputs without being the best vehicle for extrapolation.

4.2 Positioning relative to prior work

Bilinear models in machine learning. Factorized bilinear pooling [3], multimodal bilinear fusion [1], and attention-based bilinear methods [2] all address the same computational motif: capturing multiplicative interactions between two input streams. These methods are implemented digitally with learned weight matrices. Our work shows that a physical scattering medium can serve as a fixed, high-dimensional bilinear kernel whose interaction structure is determined by wave physics rather than gradient descent. The advantage is not that the physical kernel is "better trained" but that it provides a qualitatively different representation—one carrying coherent phase information inaccessible to intensity-based digital products.

Optical random features and reservoir computing. Prior optical computing work has primarily treated scattering media as single-input random-feature engines [5] or reservoir substrates [6]. Our contribution extends this paradigm to the two-input regime: by coherently superposing two token fields and demodulating the cross-term, we access a bilinear observable that goes beyond the single-input embedding.

Interferometric measurement. Phase-stepping interferometry is a mature technique in optical metrology, from X-ray phase imaging [7] to phase-sensitive reflectometry [8]. Our four-phase demodulation protocol applies the same principle to isolate a specific physical cross-term in a scattering system, rather than to image a sample. The methodological novelty is not the demodulation itself but its application to extract a computationally meaningful bilinear observable.

4.3 Limitations

We identify five concrete limitations of the present work:

1. **Scale.** The largest vocabulary tested is 8 tokens (64 ordered pairs). Whether the structural separation survives at vocabularies of hundreds or thousands of tokens remains an open question. Prior work on the same platform at $N = 16$ tokens showed that some token-family encodings suffer degeneracy-boundary collapse as pair counts grow [4]; similar effects may constrain Complex-$\boldsymbol{B}$ at larger scale.

2. **Generalization.** The cross-split experiment (section 3.4) reveals that Complex-$\boldsymbol{B}$'s category-relation signal does not generalize as cleanly as the digital bilinear baseline to unseen token pairs. This limits the current system's applicability to tasks requiring abstract relational transfer.

3. **Token-field mapping.** The semantic tokens are mapped to optical fields via a deterministic phase-only superpixel codebook. This mapping is arbitrary in the sense that "cat" and "dog" do not share more optical structure than "cat" and "hammer." The observed semantic-category separation therefore arises from the interaction physics, not from a semantically informed encoding.

4. **Classifier simplicity.** All classification uses ridge regression with leave-one-out cross-validation. More powerful classifiers might narrow or widen the gap between feature families. The structural separation (concat never solves relation, digital bilinear never solves identity) appears robust under the tested linear protocol because it involves floor-level or chance-level failures rather than marginal differences, but we have not verified this with nonlinear classifiers, and absolute accuracy values may change with different evaluation methods.

5. **Proof-of-principle scope.** This work demonstrates a physical bilinear interaction engine and maps its structural properties. It does not yet demonstrate a downstream application where this engine outperforms optimized digital pipelines on a practical task. The contribution is to the understanding of what physical scattering systems can compute, not to immediate engineering utility.

4.4 Outlook

The results suggest several directions for future work:

- **Scaling studies** at $N = 16$–64 tokens to map where the joint-encoding advantage persists and where it collapses.

- **Learned token-field mappings** that encode semantic similarity into optical-field similarity, potentially improving category-level generalization.

- **Integration with digital post-processing**: using Complex-$\boldsymbol{B}$ as a physical feature layer within a hybrid optical–digital pipeline.
- **Theoretical analysis** of the information capacity of the bilinear interaction field as a function of the transmission matrix rank and the token-field design.

5 Methods

5.1 Optical platform

All experiments were performed on a self-referenced optical scattering platform. A coherent 633 nm laser illuminated a phase-only spatial light modulator (SLM, 1920×1200 pixels). The modulated beam propagated through a fixed scattering medium and was detected by two cameras: CMOS1 for field documentation and CMOS2 for calibrated intensity measurements. A static self-reference beam co-propagated through the system, enabling interferometric demodulation without an external reference arm. The core bilinear interaction was implemented optically; digital processing was limited to calibration, demodulation, and downstream classification.

5.2 Token generation

Experiment 1: XOR showcase. Four tokens were encoded as deterministic phase-only SLM patterns.

Experiment 2: semantic benchmark. Eight semantic tokens—*cat*, *dog*, *hammer*, *saw*, *apple*, *grape*, *car*, *train*—were encoded as deterministic phase-only superpixel patterns. Each token used a unique seed $10{,}000 + \mathrm{id} \times 7919$ to generate a 6×6 grid of phase superpixels with block size 20 pixels and gap 4 pixels, yielding a direct one-token-to-one-field mapping with no intermediate random projection.

5.3 Acquisition protocol

Both experiments followed the same four-group structure.

Group 1: single-token baselines. Each token x was displayed alone and measured on CMOS2 with 5 repeats, yielding the single-token embedding $z(x)$.

Group 2: blank-control phase cycles. For each token y, the field $s + e^{i\phi}E_y$ (with $E_x = 0$) was measured at $\phi \in \{0, \pi/2, \pi, 3\pi/2\}$ with 5 repeats each, yielding $Q_m(0, y)$.

Group 3: pair interaction phase cycles. For each ordered pair (x, y), the field $s + E_x + e^{i\phi}E_y$ was measured at the same four phases with 5 repeats each. Demodulation followed by blank subtraction yielded $B_m(x, y)$.

Group 4: drift checks. A subset of tokens was remeasured at the end of the experiment to assess temporal stability.

Measurement counts. Experiment 1 consisted of 20 single-token measurements, 80 blank-control measurements, 320 pair-interaction measurements, and 12 drift measurements, totalling 432 calibrated measurements in 30.7 minutes. Experiment 2 consisted of 8 CMOS1 documentation shots, 40 single-token measurements, 160 blank-control measurements, 1,280 pair-interaction measurements, and 24 drift measurements, totalling 1,512 calibrated measurements in 93.3 minutes. Across both experiments, the total was 1,944 calibrated measurements.

5.4 Calibration and spatial binning

Each CMOS2 region of interest was 500×500 pixels. After calibration, each ROI was binned into 25×25 pixel blocks, yielding $20 \times 20 = 400$ spatial channels per sample. For the semantic benchmark, the experiment script initially reported near-zero early-versus-late drift correlations; direct reanalysis showed that within-group repeat stability remained excellent (typically 0.995–0.999 after binning), while the low end-to-end drift metric arose from adaptive exposure and power normalization changes over the 93-minute run rather than loss of the underlying interaction structure.

5.5 Four-phase demodulation and blank subtraction

For each channel m, the demodulated quantity was computed as

$$Q_m(x, y) = \frac{I_m^{(0)} - I_m^{(\pi)}}{4} + i\frac{I_m^{(3\pi/2)} - I_m^{(\pi/2)}}{4}. \tag{10}$$

The blank-control measurement yielded $Q_m(0, y)$, and the complex interaction field was isolated by

$$B_m(x, y) = Q_m(x, y) - Q_m(0, y). \tag{11}$$

The primary physical object was therefore $\boldsymbol{B}(x, y) = (B_m(x, y))_{m\in\mathcal{S}} \in \mathbb{C}^{400}$.

5.6 Real-coordinate representation

For numerical classification, the complex field was represented losslessly as

$$\rho(\boldsymbol{B}) = [\operatorname{Re}\boldsymbol{B}; \operatorname{Im}\boldsymbol{B}] \in \mathbb{R}^{800}. \tag{12}$$

This coordinate map preserves the full complex interaction field. Derived quantities such as $|\boldsymbol{B}|$, $\operatorname{Re}\boldsymbol{B}$, or $\operatorname{Im}\boldsymbol{B}$ alone were not treated as the primary object of the study.

5.7 Feature families

We evaluated the following feature families:

1. $z(x)$ only,
2. $z(y)$ only,
3. $[z(x), z(y)]$ concatenation,
4. $z(x) + z(y)$,
5. $z(x) - z(y)$,
6. digital bilinear baseline $z(x) \odot z(y)$,
7. Complex-$\boldsymbol{B}$: $[\operatorname{Re}\boldsymbol{B}; \operatorname{Im}\boldsymbol{B}]$,
8. hybrid $[z(x), z(y), \operatorname{Re}\boldsymbol{B}, \operatorname{Im}\boldsymbol{B}]$.

The digital bilinear baseline provided a matched-structure comparison using products of single-token intensity embeddings, thereby isolating the role of coherent phase information in Complex-$\boldsymbol{B}$.

5.8 Classification protocol

All classification used ridge regression (RidgeClassifier, $\alpha = 1.0$) with leave-one-out cross-validation. Features were standardized to zero mean and unit variance. Balanced accuracy was reported for all tasks to handle class imbalance.

5.9 Compressed-budget analysis

For the semantic benchmark, we additionally evaluated matched feature budgets $k \in \{10, 25, 50, 100, 200, 400\}$. For families with more than k features, k channels were selected uniformly at random and results were averaged over 10 random subspace trials. This analysis tested whether the structural separation between concatenation, digital bilinear, and Complex-$\boldsymbol{B}$ persisted under equal information budgets.

5.10 Cross-split generalization

For Task D, 16 ordered pairs (one per category-pair combination) were held out as test data, while the remaining 48 ordered pairs formed the training data. The task was binary same-category versus different-category classification. Held-out pairs were selected deterministically with random seed 42.

5.11 Data and code availability

All calibrated ROI arrays, metadata, token definitions, pair manifests, and analysis scripts were archived in the session workspace. CMOS1 documentation images were preserved for all eight semantic tokens.

References

[1] Yu Zhou, Jun Yu, Jianping Fan, and Dacheng Tao. Multi-modal factorized bilinear pooling with co-attention learning for visual question answering. *arXiv (Cornell University)*, 2017.

[2] Rina Kumari and Asif Ekbal. Amfb: Attention based multimodal factorized bilinear pooling for multimodal fake news detection. *Expert Systems with Applications*, 2021.

[3] Jin-Hwa Kim, Kyoung Woon On, Woosang Lim, Jeonghee Kim, Jung-Woo Ha, et al. Hadamard product for low-rank bilinear pooling. *arXiv (Cornell University)*, 2016.

[4] Sébastien M. Popoff, Geoffroy Lerosey, Rémi Carminati, Mathias Fink, A. C. Boccara, et al. Measuring the transmission matrix in optics: An approach to the study and control of light propagation in disordered media. *Physical Review Letters*, 2010.

[5] Alaa Saade, Francesco Caltagirone, Igor Carron, Laurent Daudet, Angélique Drémeau, et al. Random projections through multiple optical scattering: Approximating kernels at the speed of light. In *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2016.

[6] Y. Paquot, F. Duport, A. Smerieri, J. Dambre, B. Schrauwen, et al. Optoelectronic reservoir computing. *Scientific Reports*, 2012.

[7] Timm Weitkamp, Ana Díaz, Christian Dávid, Franz Pfeiffer, Marco Stampanoni, et al. X-ray phase imaging with a grating interferometer. *Optics Express*, 2005.

[8] Shuaiqi Liu, Feihong Yu, Rui Hong, Weijie Xu, Liyang Shao, et al. Advances in phase-sensitive optical time-domain reflectometry. *Opto-Electronic Advances*, 2021.